

Accuracy of a requirements ambiguity detector: case of a pet social network

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Abstract

Context: Ambiguity is one of the most persistent defects in natural-language requirements specifications, and rule-based detectors are frequently proposed as a low-cost mechanism to flag it before requirements reach design. Evidence on how closely such detectors reproduce human expert judgement remains scarce for Spanish-language specifications produced in Latin American settings.

Objective: This study measures the extent to which a rule-based ambiguity detector replicates the consensus judgement of independent human experts over the complete requirements corpus of MundiPets, a real social network for responsible pet adoption and breeding, and characterises the patterns of disagreement between the two.

Methods: We conducted a paired quasi-experiment over 61 requirements (27 functional, 16 non-functional, 9 design constraints and 9 legal requirements) specified according to an eight-attribute template. A purpose-built rule-based detector, implemented in Python and frozen before data collection, classified each requirement as ambiguous or non-ambiguous using four lexical and syntactic rules. In parallel, three independent software engineers, blind to the detector output and to one another, classified the same requirements in randomised order

using a shared rubric. The protocol was preregistered on the Open Science Framework before any data were collected. Agreement was quantified with precision, recall, F_1 , Cohen’s κ and Fleiss’ κ .

Results: The detector flagged 32.79% of requirements as ambiguous (20/61) against 26.23% (16/61) for the expert consensus, yielding accuracy = 0.8033 (95% CI [0.7049, 0.9016]), precision = 0.600, recall = 0.750 and $F_1 = 0.667$ for the ambiguous class. Agreement between detector and consensus was moderate (Cohen’s $\kappa = 0.530$, 95% CI [0.280, 0.737]) and a chi-squared test of independence found the detector significantly associated with expert consensus beyond chance ($\chi^2 = 15.037$, d.f. = 1, $p = 0.0001$, $\phi = 0.497$). The expert panel itself agreed almost perfectly (Fleiss’ $\kappa = 0.857$), which strengthens confidence in the reference standard the detector is compared against. Coincidence with expert consensus was strongest for legal requirements (9/9, 100%) and non-functional requirements (14/16, 87.5%), and weakest for design constraints (5/9, 55.6%).

Conclusion: A four-rule lexical-syntactic detector achieves moderate, statistically significant agreement with an almost-perfectly agreeing expert panel on a real, Spanish-language requirements corpus, performing best on non-functional and legal requirements and worst on design constraints, where ambiguity is more often a matter of missing procedural detail than of detectable lexical patterns. The result supports using such a detector as a first-pass triage tool in a review workflow, while showing that design constraints in particular still require human judgement.

Keywords: Requirements Engineering, Empirical Software Engineering, Natural Language Processing, Interrater Agreement, Requirements Quality

1 Introduction

MundiPets is a social network for responsible pet adoption and breeding, currently under specification for veterinary practices and pet owners in the province of Los Ríos, Ecuador. The system lets owners publish verified pet profiles, lets registered veterinary practices validate the associated medical records, and supports adoption and breeding requests through a staged workflow that includes compatibility assessment by an artificial-intelligence component. Requirements engineering matters acutely in this domain because a defect in a requirement about medical-record validation or about the visibility of an owner’s location does not merely degrade usability: it propagates into a system that handles animal health data and personal data protected under Ecuador’s Organic Law on Personal Data Protection [?]. Specifying such a system with verifiable, unambiguous requirements is therefore a precondition for lawful operation rather than a stylistic preference.

Ambiguity remains one of the defects most consistently reported in natural-language requirements. The systematic mapping study of Montgomery et al. [?] shows that, although requirements quality has been studied extensively, empirical work concentrates on a small number of quality attributes and rarely validates automated detection against human judgement on complete, real specifications. Atoum et al. [?]

reach a convergent conclusion in their systematic review of requirements quality assurance, identifying the scarcity of validated automated support as an open challenge. Ferrari et al. [?] demonstrated that ambiguity in elicitation is entangled with tacit knowledge, which suggests that purely lexical detection has an intrinsic ceiling; yet the size of that ceiling has not been quantified for specifications written in Spanish, nor for corpora produced by novice analysts, the setting in which a large share of requirements documents are in fact written. This is the gap the present study addresses.

This paper makes three contributions, each grounded in evidence collected by the authors and released as an open dataset:

1. a quantified comparison between a rule-based ambiguity detector and the blind consensus of three independent human experts over a complete requirements corpus of 61 items — functional requirements, non-functional requirements, design constraints and legal requirements — written in Spanish;
2. a characterisation of the disagreement patterns between automated and human classification, reported per requirement and per rule, which exposes the specific linguistic phenomena that a rule-based approach systematically over- or under-flags;
3. a replication package — corpus, detector output, expert classifications, analysis scripts and preregistered protocol — released under CC BY 4.0 so that the study can be repeated on other corpora and other languages.

The study answers two research questions:

- RQ1** With what precision, recall and F_1 does a rule-based automatic detector replicate the consensus judgement of human experts when classifying the requirements of MundiPets as ambiguous?
- RQ2** Which disagreement patterns predominate between the automatic detector and expert judgement, and what do they reveal about the limits of a rule-based approach relative to human judgement in identifying ambiguity?

The remainder of the paper is organised as follows. Section 2 positions the study within the closest related work. Section ?? describes the experimental design, participants, instruments and analysis plan. Section ?? reports the results, Section ?? interprets them, Section ?? discusses threats to validity, and Section ?? concludes.

2 Related work

This section is not a full systematic literature review; it follows the abbreviated protocol of Kitchenham and Charters [?], reporting a documented search string, explicit inclusion and exclusion criteria, the number of studies retrieved and screened, and a comparative table of the closest works.

2.1 Search strategy

The search string combined three concept blocks:

("requirements engineering" OR "software requirements") AND ("ambiguity" OR "requirements smell" OR "requirements quality") AND ("automatic detection" OR "rule-based" OR "natural language processing")

The string was executed on Scopus and Google Scholar on 2 August 2026, restricted to the 2023–2026 period. Inclusion criteria: peer-reviewed studies reporting empirical data on the quality, ambiguity or automated analysis of natural-language requirements, published in requirements-engineering or software-engineering venues. Exclusion criteria: editorials, short contributions under four pages, and studies without empirical data. The search retrieved 689 records in Scopus and 3,670 in Google Scholar. After title and abstract screening, 30 records were read in full text and 6 were retained. These were combined with eight studies already identified in the project bibliography, yielding the 14 entries compared in Table 1.

2.2 Comparison with the closest studies

Table 1 contrasts the retained studies with the present work. The first eight rows correspond to studies already identified in the project bibliography; the last six were retained from the search described above.

The niche of the present study is narrow and, to our knowledge, unoccupied: a paired comparison between a frozen, fully documented rule-based detector and the blind consensus of independent experts, over the *complete* requirements corpus of one real system, specified in Spanish, with the corpus, the detector output and the individual expert classifications released as open data. Existing work either maps the field without primary agreement data, or evaluates detectors on English-language benchmark corpora that were not produced by the team reporting the evaluation.

Table 1 Comparison with the closest related studies

Reference	Year	Study type	Population	Difference with our work
Montgomery et al. [?]	2022	Systematic mapping	Primary studies on requirements quality	Maps the field but reports no primary comparison between an automated detector and expert consensus on a single complete corpus.
Atoum et al. [?]	2021	Systematic review	Literature on requirements quality assurance and validation	Identifies the lack of validated automation as an open challenge; does not measure detector-expert agreement empirically.
Ferrari et al. [?]	2016	Empirical study	Elicitation interviews	Studies ambiguity as it arises in elicitation and its link to tacit knowledge; our unit of analysis is the written requirement in a closed specification.
Fischbach et al. [?]	2023	Tool evaluation	Conditional statements in natural-language requirements	Derives acceptance tests automatically; shares the rule/NLP approach but targets test generation, not ambiguity classification.
Gobov [?]	2023	Practitioner survey	Specification and modelling techniques in practice	Characterises which techniques practitioners use; provides no automated quality measurement.
Said et al. [?]	2026	Model evaluation	SRS documents	Classifies requirements as functional or non-functional with a learned model; our detector targets ambiguity and is rule-based and fully auditable.
Cheng et al. [?]	2026	Systematic review	Generative AI applied to requirements engineering	Surveys generative approaches; we deliberately avoid large language models to keep every classification traceable to a named rule.
Vogelsang and Fischbach [?]	2024	Methodological guideline	LLM use for NLP tasks in RE	Proposes usage guidelines rather than reporting an agreement study.
Abdeahad et al. [?]	2026	Model evaluation	7,061 requirements from the Fault-Prone SRS benchmark	Achieves high accuracy with a deep model, but evaluates against dataset labels rather than against a blind expert panel, and the model is not auditable rule by rule.
Fantechi et al. [?]	2023	Tool comparison	Three example requirements documents	Closest to our design: compares a rule-based tool with an LLM on ambiguity. Uses the authors themselves as the reference judgement, whereas our reference is a blind panel independent of the specification team.
Talha et al. [?]	2025	Tool evaluation	1,553 requirements across 17 domains 5	Targets anaphoric and coordination ambiguity with a transformer; our detector targets lexical and syntactic smells and is compared against human consensus rather than a labelled benchmark.
Alemneh and Berhanu [?]	2024	Systematic review	Literature on requirements smells and detection techniques	Catalogues detection techniques; reports no primary agreement measurement on a real corpus.
Intana et al. [?]	2024	Tool evaluation	Thai-language SRS documents	The closest non-English precedent; confirms that language-specific evaluation is needed but does not cover Spanish nor report inter-rater agreement.
Bashir et al. [?]	2025	Industrial study	Three railway requirements datasets	Reports the over-flagging pattern (high recall, low precision) also observed in rule-based tools; uses LLMs, which we deliberately avoid for auditability.